

Lab 5: Introduction to pavo — Quantifying Reef Fish Color Patterns

Week 6

EEB 187 — Ecology & Evolution of Color

Overview

This is the first computational analysis lab of the course. You will use the R package **pavo** to extract color pattern metrics from reef fish photographs.

Format: Instructor-led demo on three reef fish from three different families + independent practice on three fish from your group's assigned lineage.

- **Demo + walkthrough** (instructor + Rosamari): 35–40 min
- **Independent practice** (3 fish from your lineage): 30–35 min
- **Wrap-up + cross-lineage comparison:** 5–10 min

Bring:

- Laptop with R, RStudio, and pavo installed (see `pre-lab-installation.qmd`)
- The lab bundle: `labs/demo-pavo-week6/` (download from BruinLearn)
- Three image files from your assigned lineage saved locally before lab — see Part B Step 1

Part A: The demo — three reef fish, three pattern axes

We're starting with three fish from three different families. Each fish has a fundamentally different pattern type. The pedagogical claim of the lab: pavo's adjacency framework can distinguish these patterns in numbers — specifically with the **A statistic** (aspect ratio).

Demo fish	Family	Pattern	Predicted A
Sergeant major (<i>Abudefduf saxatilis</i>)	Pomacentridae	Vertical bars	$A > 1$
Cleaner wrasse (<i>Labroides dimidiatus</i>)	Labridae	Horizontal stripe	$A < 1$
Mandarin fish (<i>Synchiropus splendidus</i>)	Callionymidae	Reticulate maze	$A \approx 1$

Follow along with `scripts/walkthrough.R`. Rosamari is presenting the slide deck; the R script is on your laptop.

Record the demo-fish metrics here for reference later:

Metric	Sergeant major	Cleaner wrasse	Mandarin fish
k			
m			
A			
Jc			
Jt			
m_dS			
m_dL			

Part B: Your turn — three fish from your assigned lineage

Your group has been assigned a reef fish lineage (typically a family). Today's job: pick **three species from your lineage** that show **three visually different pattern types**. Run all three through the pavo pipeline. Compare your results to the demo fish from Part A.

Step 1: Pick three fish

i Where to find images depending on your launch path

Zero-hassle option (works in every launch path): the repo ships with 5 species per assigned lineage in `images/student-fish/<your-lineage>/`. Pick 3 of those and you're done — no uploading, no internet hunting.

If you want to use a fish that isn't pre-loaded, **put it in the dedicated `images/student-images/` folder** (kept separate from the curated starter library so it's clear what's yours):

- **Docker / Native:** drop your JPG into `~/eeb187-pavo-lab/images/student-images/` on your laptop. The bind mount makes it visible inside the R session immediately. Refer to it as `"images/student-images/my-fish.jpg"`.
- **Binder (cloud, ephemeral):** in RStudio's *Files* pane (bottom-right), navigate to `images/student-images/`, click **Upload** → pick your JPG → OK. Refer to it as `"images/student-images/my-fish.jpg"`. **Binder sessions expire after 90 min — finish in one sitting**, or download segmented outputs before logging off.
- **Codespaces (cloud, persistent):** drag-drop the JPG into the VS Code file explorer at `images/student-images/`. Files persist between visits.

Browse the available images for your assigned lineage. (FishView, FishBase, your group's curated set, or `images/student-fish/<your-lineage>/` which is pre-loaded for every launch path.) Pick three species that look as different as you can find — for example:

- One with strong **vertical bars** (you predict $A > 1$)
- One with a **horizontal element** like a stripe or band (you predict $A < 1$)
- One that's **spotty, reticulate, or solid** (you predict $A \approx 1$)

If your lineage doesn't have all three pattern types, pick the three that span the widest pattern range available.

Write your three species here:

Lineage: _____

Fish 1 (predict $A > 1$): _____

Fish 2 (predict $A < 1$): _____

Fish 3 (predict $A \approx 1$): _____

Step 1.5: SEGMENT — remove the background from each image

This is the most important pre-processing step. Without it, pavo's `k-means classify()` runs over the WHOLE image including water, reef, and tank glass. Background pixels become “color classes,” and the resulting metrics describe the photograph rather than the fish.

The 3 demo fish you saw in Part A were segmented using **rembg**, an AI background-removal tool. The demo's dramatic A values (1.54 / 0.60 / 1.02) depend on this segmentation step. You have two options:

Path 1 (recommended): rembg via segment_fish() R helper

The `segment_fish()` function in `walkthrough.R` wraps `rembg` + `ImageMagick` into one R call. **One-time setup** (you should have done this in pre-lab Step 6):

```
# Install uv (handles rembg's Python + dependencies automatically):
brew install uv                # macOS
curl -LsSf https://astral.sh/uv/install.sh | sh    # Linux
# Plus ImageMagick:
brew install imagemagick       # macOS
sudo apt install imagemagick   # Linux
```

Per-fish use (in R during lab):

```
# Use the same relative path students see in the Files pane -
# images/student-images/<filename>.jpg for fish you uploaded yourself,
# images/student-fish/<lineage>/<filename>.jpg for the pre-loaded library.
seg_path <- segment_fish("images/student-images/your-fish.jpg")
my_fish  <- getimg(seg_path)
```

The result is a JPG with the fish on a clean white background, saved as `<name>-segmented.jpg`. If `segment_fish()` errors out, fall back to Path 2.

 If `segment_fish()` fails with “rembg failed (or not installed)”

R may not see your shell's PATH. First confirm the tools work in your terminal:

```
uvx --from "rembg[cpu,cli]" rembg --help
magick -version
```

If both work in terminal but `segment_fish()` still fails, R's PATH is incomplete. Check it:

```
Sys.getenv("PATH")
```

If `/opt/homebrew/bin` (Homebrew on macOS) or `~/local/bin` isn't there, add it to `~/Renviron`:

```
PATH=/opt/homebrew/bin:/usr/local/bin:/usr/bin:/bin:~/local/bin
```

...and restart RStudio. If you're still stuck, use Path 2 (web tool) or Path 3 (manual crop) below — both work without needing R to find any shell tools.

Path 2 (no install, web-based): segment with [remove.bg](#) or [photoroom.com](#)

Drag-drop your fish JPG into one of these free tools:

- [remove.bg](#) — fastest, no signup
- [photoroom.com](#) — also free
- [clipdrop.co/remove-background](#) — Stability-AI tool

Each returns a transparent PNG. Save the PNG into the `images/student-images/` folder (your personal upload area). Then in R, use the `composite_to_white()` helper to add a white background and standardize the canvas:

```
source("scripts/helpers.R") # loads composite_to_white()
seg_path <- composite_to_white("images/student-images/your-fish.png")
my_fish <- getimg(seg_path)
```

This gives you the same `*-segmented.jpg` you'd get from `segment_fish()` — pavo's classify-with-bkg path will work exactly the same way.

Path 3 (fallback, no install, no internet): manual crop

Open each fish image in **Preview.app** (or any image editor), drag a selection rectangle as tight to the fish as possible without cutting any fins, then **Tools** → **Crop** and save. Load with `getimg("path/to/cropped.jpg")` — skip both `segment_fish()` and `composite_to_white()`.

💡 Tradeoff for the manual-crop path (Path 3)

Cropped images still have water/reef in the corners. **Your A values will be less dramatic than the demo** (closer to $A=1$ for all three) because some background pixels will pollute the k-means classification and the adjacency grid. The directional ordering still holds — vertical-bar fish > 1 , horizontal-stripe fish < 1 — just with smaller effect

sizes. Note in your worksheet which path you used. The rubric accommodates all three. Step 2 below has a different code block for the manual-crop path: drop the `bkgID = my_bkg`, `exclude = "background"` arguments since there's no white background class to exclude. (Paths 1 and 2 both produce a white-background JPG and use the standard code block.)

Comparing the three folders for the demo fish

Look at `images/demo/raw/` (uncropped originals) and `images/demo/segmented/` (rembg output) to see exactly what segmentation does. Run `getimg()` on both and compare the resulting `classRGB` palettes — the raw ones include a “blue water” class; the segmented ones don't.

Step 1.6: REGISTER each image to left-lateral view

Standard ichthyology convention: head on the LEFT, tail on the RIGHT. Before classifying, look at each of your three images and confirm the head is on the left. If a fish is facing the wrong way, flip it horizontally:

```
# Defined in walkthrough.R
flip_to_left_lateral("path/to/my_fish.jpg")
```

Or from a terminal (macOS / Linux with ImageMagick installed):

```
magick my_fish.jpg -flop my_fish.jpg
```

This step matters because:

- Predictions like “vertical bars” presume the canonical orientation
- Cross-fish comparison in the next pavo session expects all images registered the same way
- Same goes for any biology paper you'll publish with this kind of analysis

Note: horizontal mirroring does NOT change the A, m, Jc, or other adjacency metrics — they're invariant to left/right reflection. Registration is for visual consistency and comparative validity, not metric correctness.

Step 2: Run the pipeline on each fish

For each of your three segmented + registered fish, run:

```

my_fish <- getimg(seg_path)          # path returned by segment_fish() or your crop
plot(my_fish)

# k = (number of distinct fish colors) + 1 if you used rembg (the +1 is for
# the white background). For manual-crop path, just the fish-color count.
my_k    <- 4
my_class <- classify(my_fish, kcols = my_k)
plot(my_class)

# Identify and exclude white background (only for rembg path)
my_bkg <- pick_white_bg(my_class)   # = which.max(rowSums(classRGB))
my_cd  <- simple_coldists(my_class)
my_adj <- adjacent(my_class, xpts = 200, xscale = 5, coldists = my_cd,
                   bkgID = my_bkg, exclude = "background")
# For the manual-crop path: drop the bkgID + exclude args.

# Pull the metrics
my_metrics <- tibble(
  k    = pick_col(my_adj, c("k", "kcols")),
  m    = pick_col(my_adj, c("m", "m_r")),
  A    = pick_col(my_adj, c("A", "Asp")),
  Jc   = pick_col(my_adj, c("Jc")),
  Jt   = pick_col(my_adj, c("Jt")),
  m_dS = pick_col(my_adj, c("m_dS", "Cs_dS")),
  m_dL = pick_col(my_adj, c("m_dL", "Cs_dL")),
)

```

! Load the helpers FIRST

All the helper functions (`pick_col()`, `simple_coldists()`, `pick_white_bg()`, `segment_fish()`, `composite_to_white()`, `flip_to_left_lateral()`) live in `scripts/helpers.R`. Run this **once at the top of your session** before anything else:

```
source("scripts/helpers.R")
```

If you skip this and try to call e.g. `simple_coldists(my_class)`, you'll get **Error: could not find function "simple_coldists"**. Source it once and every helper is in scope for the rest of your session.

Step 3: Record your results

Metric	Fish 1	Fish 2	Fish 3
Species			
k			
m			
A			
Jc			
Jt			
m_dS			
m_dL			

Step 4: Compare to the demo fish

Look at your three fish's A values vs. the three demo fish (Sergeant major, Cleaner wrasse, Mandarin) from Part A.

(a) Did the A statistic correctly classify your three fish by pattern orientation?

_____ YES — my barred fish has $A > 1$, my horizontal fish has $A < 1$, my reticulate/spotty fish has $A \approx 1$

_____ PARTIALLY — explain which one(s) didn't behave as expected:

_____ NO — none of them landed where I predicted. Speculate why:

Notes:

(b) Cross-lineage comparison. Pick ONE of your three fish whose pattern most resembles one of the demo fish. Compare their full metric profiles (m, A, Jc, m_dL).

My fish: _____

Most similar demo fish: _____

What's similar in the metrics:

What's different in the metrics:

Step 5: Biological interpretation

In 4–6 sentences, answer:

(a) **Within your lineage:** are the three fish you picked more similar to each other in metric space than they are to fish from other families? Or are some of your three fish a closer metric match to the demo fish than to each other? What might that suggest about pattern evolution within vs. across reef fish lineages?

(b) **Selection hypothesis:** for ONE of your three fish, propose a selection pressure that might have favored its pattern. Pick from: crypsis, aposematism, species recognition, sexual signaling, predator deflection (eyespot), disruptive coloration, or “I have a hypothesis but I’m uncertain — here’s why.”

Submitting

Submit by end of day to BruinLearn. Either:

- A rendered HTML or PDF of this .qmd with your tables filled in, OR
- A 1-page Word/Google doc with the **3 metric tables** + the **comparison answers** + the **biological interpretation**.

The R code does not need to be polished. The biological reasoning is the assessable part.

What's next

There will be a **second pavo session** later in the term. We'll build a metric matrix across many species per lineage, cluster fish by pattern, and compare the clusters to phylogeny. Today's lab gives you the tool; next time we use it to ask comparative questions.

Common errors and their fixes

Error in getimg: cannot open the connection — file path is wrong. Use `getwd()` to see where R thinks you are. In RStudio: Session → Set Working Directory → To Source File Location.

Classify image looks weird (washed out, only 2 colors) — try a different `k`. If your fish has 5 visible color regions, `k = 5`.

Error: object 'classify' not found — you didn't run `library(pavo)`. Run it.

Adjacency runs but everything is NA — your image is empty / all-white / all-black. Re-load with `getimg()` and check `plot(img)` actually shows the fish.

The metrics table looks empty in some columns — `pavo` column names vary by version. The `pick_col()` helper in `walkthrough.R` falls back gracefully; if any metric is `NA`, look at `colnames(my_adj)` to find the actual name in your version.

My A values for the 3 fish don't match my predictions — this is real! Common causes: (1) `k` is too small or too large, collapsing or splitting biological color regions; (2) the image background isn't well-segmented (background bleeds into the analysis); (3) the fish actually doesn't have the pattern you thought it had — `pavo` is a more sensitive eye than yours. Note this in your interpretation; it's good science.

What this lab is NOT

We are not (yet) doing:

- Spectral analysis from a spectrometer (different `pavo` functions: `procspec`, `peakshape`, etc.)
- Visual modeling through a fish's eyes (`vismodel`, `coldist`, JNDs)
- Many-species comparative analysis (next `pavo` session)
- Phylogenetic comparative methods on color (Lab 9, end of term)

Today we extract pattern metrics from three reef fish, predict-and-test, and compare across lineages. That alone is a substantive analytical exercise.